

Hawks, Doves, and the Dot Plot: Heterogeneous FOMC Reaction Functions*

Vladislav Abramov[†]

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Abstract

This paper analyzes the individual policy reaction functions of Federal Open Market Committee (FOMC) members using the Summary of Economic Projections data from 2012 to 2019. I estimate member-specific policy reaction functions and document substantial heterogeneity in the marginal effects of inflation and unemployment gaps across members. I show that this heterogeneity implies differences in voting behavior and forecasted policy rate paths. Observable individual characteristics—such as educational background, age, and research experience—account for a significant share of the variation. These findings point to systematic differences in policy preferences among FOMC participants.

Keywords: monetary policy rules, FOMC members' preferences, dissent voting

JEL Classification: E52, E58

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[†]Department of Economics, Texas A&M University, 2935 Research Parkway Suite 200, College Station, TX 77843, United States. E-mail: vabramov@tamu.edu.

1 Introduction

Monetary policy in the United States is conducted by the Federal Open Market Committee (FOMC), a body composed of Board of Governors members and regional Federal Reserve Bank presidents. There are eight regular policy meetings per year, and twelve voting members who make policy decisions. While the FOMC operates under a shared mandate to pursue price stability and maximum employment, each FOMC member brings unique experiences, educational and professional backgrounds that can influence how aggressively—or cautiously—they respond to changes in inflation or labor market conditions. How different policymakers perceive the trade-off between inflation and unemployment stabilization can affect internal discussions and voting, ultimately influencing the Committee’s collective decisions.

Given this diversity of perspectives, a natural question arises: to what extent do individual FOMC members differ in their policy stances, and how do these differences translate into their policy decisions? This question remains relatively understudied, in part because there is limited empirical evidence on the extent to which individual views differ within the committee, see Istrefi (2019) and Bordo and Istrefi (2023) for recent exceptions. A key challenge is that these policy preferences are not directly observable and often difficult to infer from the data. The main objective of this paper is to assess the individual reaction functions of FOMC members—and specifically, identify the relative importance members place on inflation and unemployment deviations from their long-run targets within a linear class of rules.¹

In principle, one can characterize FOMC members’ policy stances by estimating policy reaction functions, in the spirit of Taylor (1993). Despite a substantial body of literature dedicated to monetary policy rules and their implications for aggregate economic outcomes, research on the heterogeneity within these rules, particularly at the individual member level, remains sparse. This paper contributes to the literature by estimating individual-specific policy rules for FOMC members using publicly available Summary of Economic Projections (SEP) data.

The SEP includes projections submitted by both voting and non-voting members of the FOMC for several variables and forecast horizons. Variables include headline and core inflation, real GDP growth, unemployment rate, and federal funds rate. The data include both finite horizon and long-run projections, which allow us to construct inflation and unemployment gaps forecasted by individual members. SEP forecasts

¹In this paper, I assume that FOMC members assign different relative weights to inflation versus unemployment stabilization. That is, there exists a trade-off between the two objectives, and how each member perceives and navigates this trade-off reflects their individual policy stance.

are widely used by market analysts, policymakers, and academics to assess the Fed’s macroeconomic outlook and possible course of future monetary policy. The first SEP forecasts were published after the Fed’s October 2007 meeting. Since then, the Fed has regarded SEP data as an important component of its forward guidance policy.² While not binding, these forecasts reveal participants’ current views on the policy path and the macroeconomic outlook. I merge these forecasts with the FOMC participants’ keys, which are published with a 5-year lag, yielding a panel dataset with individual member forecasts across time and forecast horizons for the 2007-2019 period.

Using this panel, I estimate policy rules for 23 FOMC members.³ I document important differences in how FOMC members view the inflation versus unemployment trade-off. Average marginal effects (AMEs) on the inflation and unemployment gaps in the estimated individual policy rules vary substantially across FOMC members—including within the same Reserve Bank district and among members of the Board of Governors. For instance, a 1 percentage point increase in the inflation gap forecast is associated with a 0 to 3.4 percentage points increase in the projected federal funds rate across members, while a 1 percentage point increase in the unemployment gap forecast corresponds to a 0.3 to 1.9 percentage points decrease. This heterogeneity is robust across estimation methods and does not hinge on any single member or small subset of members.

I summarize each member’s policy rule with a one-dimensional continuous index capturing the perceived inflation–unemployment trade-off implied by my estimates. This measure is the sum of average marginal effects of inflation and unemployment gaps from the estimated individual policy reaction functions. In contrast to the classification of members’ policy stance in Istrefi (2019), I do not classify members into separate groups (hawks, doves, and swingers) nor rely on textual analysis of news outlets. Instead, I use FOMC members’ economic projections to construct a similar classification. While I do not formally categorize members as hawks and doves in this paper, I provide a spectrum that defines the relative importance of inflation over unemployment for a given sample of FOMC members.⁴ I document that this measure of perceived tradeoff varies considerably across FOMC members.

I acknowledge that this measure may also be subject to the influence of other members’ views, economic environment, and other factors unrelated to true intrinsic preferences of the members. For example, recent research shows that regional conditions

²<https://www.federalreserve.gov/newsevents/speech/bernanke20071114a.htm>

³Though a total of 43 FOMC members served between 2007 and 2019, many members served only brief terms.

⁴My work is complementary to, but distinct from, approaches that infer objectives from policymakers’ sentiment Shapiro and Wilson (2022) and that examine perceived monetary policy rule of the Fed using micro data on professional forecasters Bauer, Pflueger, and Sunderam (2024).

affect the voting behavior of regional presidents and final monetary policy decisions (Bobrov, Kamdar, and Ulate 2024, Fos and Xu 2025). I discuss whether it is possible to interpret this measure as a measure of preferences in greater detail in Section 4. A full causal decomposition of intrinsic preferences from institutional and regional influences is beyond the paper’s scope.

This heterogeneity in policy rules implies markedly different federal funds rate forecasts, holding constant other inputs to the policy rules. To show that, I consider several counterfactual structures of the FOMC. For instance, had all FOMC members shared the Chair’s reaction function (yet still had their own distinct forecasts of inflation and unemployment), the forecasted federal funds rate would be different from the actual consensus forecast by 25-75 b.p. in most of the periods in the sample. I also found that had Jerome Powell shared the same reaction function with Janet Yellen, the forecasted policy rate by him would be different by 25-50 b.p. in most of the periods compared with forecasts implied by his reaction function (and vice versa). This analysis means that not only is the forecasted macroeconomic outlook (policy rule inputs) crucial for explaining policy decisions, but the variation in FOMC members’ reaction functions also plays a significant role.

Next, I ask the following question: can observed individual characteristics of these members explain the differences in their reaction functions? This question is not new, with several key papers approaching it with different types of data and methods. Malmendier, Nagel, and Yan (2021) show that personal experiences of inflation strongly influence the hawkish or dovish leanings of central bankers. Using the classification of FOMC members by Istrefi (2019),⁵ Bordo and Istrefi (2023) demonstrates that FOMC members’ policy stances are shaped by their birth period (and are related to experienced inflation and unemployment) and education. Hawks often studied at “freshwater” universities, while doves are more likely to be “saltwater” graduates and born during high-unemployment periods.⁶ Smales and Apergis (2016) consider a set of individual characteristics and show that decisions of policymakers are primarily associated with their tenure within the Federal Reserve System and the FOMC.

Following this literature, I consider a set of factors, including position (Board member or Reserve Bank president), voting status, working experience at the Fed, education,

⁵Istrefi (2019) creates a measure of policy preferences by classifying FOMC members into hawks (who put more emphasis on inflation), doves (who put more emphasis on economic activity), and swingers (who do not have a strong inclination). They show that this measure is related to the preferred rates of members and helps to predict dissenting votes.

⁶“Freshwater” and “saltwater” schools refer to differing macroeconomic traditions, with the former emphasizing market efficiency and policy rules (e.g., University of Chicago, University of Minnesota) and the latter favoring Keynesian approaches and active stabilization policy (e.g., Massachusetts Institute of Technology, Harvard University).

and research background of each of the FOMC members in the sample. Though there are many possible factors contributing to this heterogeneity, I mainly focus on those considered in the literature. I show that a relatively small set of observable exogenous characteristics can explain a substantial share of the heterogeneity in FOMC members' reaction functions.

Finally, I explore the role of heterogeneity in reaction functions for the actual voting records. Specifically, I look at how dissent voting is related to the measure of perceived tradeoff between inflation and unemployment. The rationale for dissent has been investigated since the early 1990s, see Belden (1991) and Havrilesky and Gildea (1991), with authors usually assuming that dissent voting is likely a result of inherent preferences for either tighter or looser monetary policy. There is also reason to believe that dissent may arise from strategic, political, or regional considerations, which need not always align with optimal policy from the standpoint of the central bank's objective function (Meade and Stasavage 2008, Bobrov, Kamdar, and Ulate 2024, Tillmann 2011). Also, the loss function of policymakers can include additional terms, reflecting material costs or benefits from overshooting or undershooting their long-run inflation and unemployment goals, Kocherlakota (2025).

I find strong evidence that dissent voting is systematically associated with parameters of individual policy reaction functions. I show that controlling for the individual characteristics of the members and time fixed effects, differences in reaction functions can predict dissent voting. The probability of being a hawkish dissenting voter, that is, the probability of voting for a tighter monetary policy compared to the consensus, is significantly higher for members with more inflation-oriented policy rules. Likewise, the probability of being a dove dissent is lower for such members.

Taken together, these patterns validate my approach: SEP-based reaction functions contain meaningful information about behavior of FOMC members. A better understanding of FOMC members' reaction functions and the factors shaping them may help researchers and market participants anticipate policy shifts under evolving leadership and changing committee composition. Such insights could also illuminate the internal dynamics of policy deliberations, ultimately leading to a deeper understanding of how monetary policy is debated, decided, and implemented in practice.

For instance, one implication of the findings is their potential to inform assessments of prospective candidates for the Federal Reserve Chair. Recently, President Trump's team reportedly began reviewing contenders to replace Chair Jerome Powell when his term ends in May 2026.⁷ While our estimates cannot speak to current front-

⁷<https://www.reuters.com/business/finance/bessent-leads-broad-search-fed-chair-t>

runners—Kevin Hassett has never served on the FOMC, and Kevin Warsh’s tenure (2006–2011) predated the inclusion of federal funds rate projections in the SEP—they can shed light on other potential nominees. For example, James Bullard, former President of the Federal Reserve Bank of St. Louis, appears in our data as less hawkish than most other regional presidents, placing similar weight on inflation and unemployment objectives. If appointed, his estimated reaction function suggests a stance broadly similar to Chair Powell’s.

The remainder of the paper is organized as follows. Section 2 describes the data used in the empirical analysis. Section 3 presents the estimation procedure for individual policy rules and documents heterogeneity in reaction functions. Section 4 discusses the measure of the perceived tradeoff between inflation and economic activity. Section 5 concludes.

2 Data

The Summary of Economic Projections (SEP) is a relatively recent data source, with the first release dating back to October 2007. It is the only comprehensive dataset that provides individual projections from Federal Reserve policymakers, covering multiple economic variables and forecast horizons. Because of this, the SEP is widely followed by analysts and financial markets as a key indicator of the FOMC’s policy outlook and macroeconomic expectations.

The most recent SEP publications do not identify individual contributors; instead, they present the distribution of forecasts across participants without revealing names or affiliations. The Federal Reserve releases participant identifiers with a considerable lag. At the time of writing, individual-level SEP data are available for the periods 2007–2014 and 2016–2019. Data for 2015 remain unavailable due to the Federal Reserve’s publication lag policy, linked to stricter disclosure rules introduced following the Global Financial Crisis. The data structure includes the FOMC member’s name along with forecasts of the federal funds rate, headline and core inflation, real GDP growth, and the unemployment rate. With the exception of long-term projections, all forecasts are made for the fourth quarter of corresponding year. Typically, forecasts are provided for five horizons: the fourth quarter of the current year t , and the fourth quarter of years $t + 1$, $t + 2$, $t + 3$, and the long run.

The SEP dataset is an unbalanced panel. Although the sample began relatively recently, in 2007, only 2 out of 43 FOMC members were present throughout the entire

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sample period. Each year, some members leave the committee as their terms end, and new members replace them. Additionally, the annual rotation of voting rights among regional Federal Reserve Bank presidents contributes further to the panel’s unbalanced nature. As a result, many policymakers have relatively few observations. Because estimating FOMC reaction functions relies on individual-level data, these limitations may lead to imprecise or unstable estimates.

To address this issue, I require each forecaster to have at least 40 time-horizon pairs of forecasts, see Figure 1. This corresponds to roughly two years of forecasts per member. Applying this requirement reduces the sample to 23 members. Some members were present in the FOMC for just a couple of meetings, and thus, I do not have a sufficient number of forecasts for them. To ensure stable and interpretable estimates of individual policy rules, I impose a minimum threshold of 40 observations per member. This balances the trade-off between sample size and estimation precision, retaining most members while maintaining a sufficient number of data points for robust inference.⁸

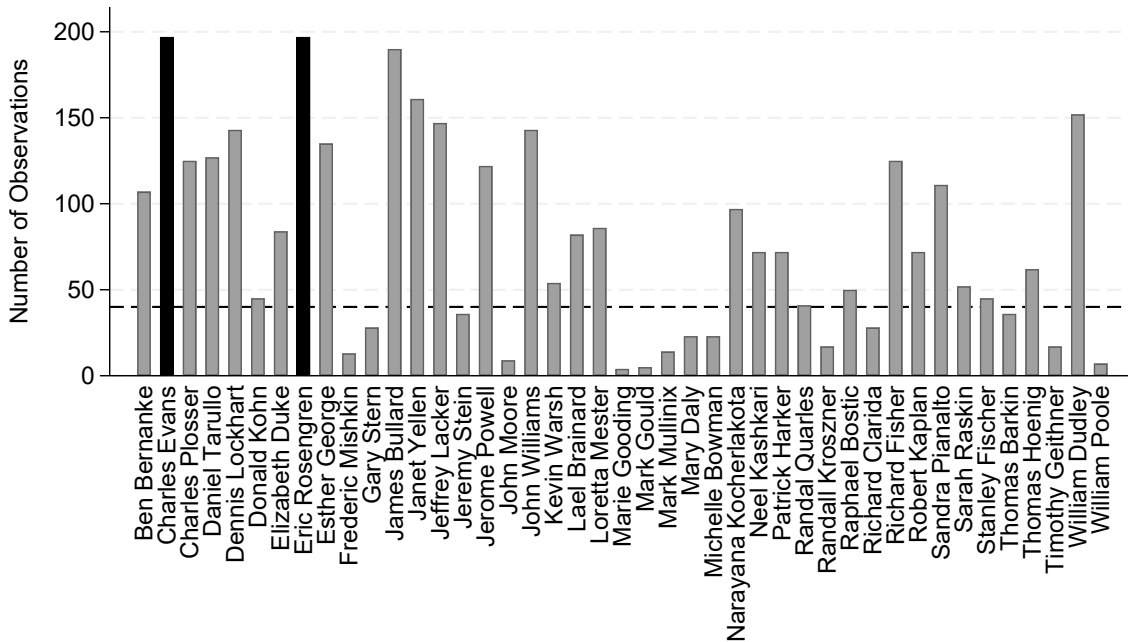


Figure 1: Sample Coverage: Number of Observations for Each FOMC Member

Figure 2 displays individual projections of the federal funds rate — the interest rate on overnight lending between depository institutions and the Fed’s primary tool for managing short-term interest rates in the economy. Popular among financial ana-

⁸Thresholds of 35, 45, and 50 observations produce similar results.

lysts, interest rate forecasts — commonly referred to as “dot plots” due to their visual presentation — were first included in SEP publications in 2012 as part of the Fed’s broader effort to increase transparency and provide more forward guidance to the public and financial markets.⁹ For much of the early sample, the federal funds rate was at or near zero, which limited the scope for traditional monetary easing. Consequently, short-term interest rate forecasts were often near zero, as policymakers believed in the necessity of maintaining an accommodative policy stance. Approximately 30% of all short-term forecasts in the sample are exactly at the zero lower bound (ZLB). This structural feature of the data—where a large share of forecasts cluster at the lower bound—underscores the need for a censored modeling approach, which I implement in the next section.

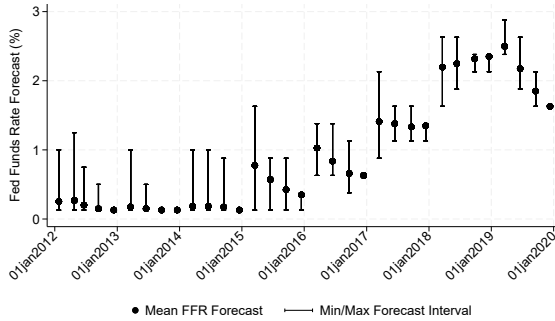


Figure 2: Federal funds Rate Forecasts

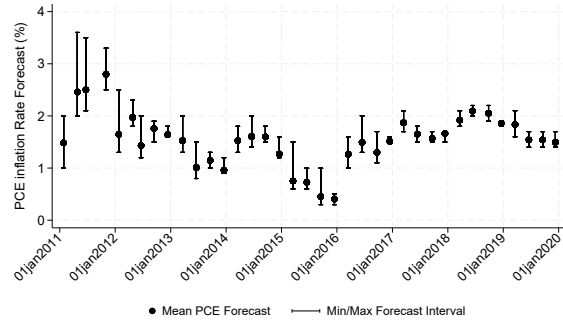


Figure 3: PCE Inflation Rate Forecasts

3 Reaction functions of FOMC members

3.1 Empirical specification

Individual reaction functions can be described by a policy rule that links the policy rate (federal funds rate) to macroeconomic variables in the Fed’s mandate, Taylor (1993). In this paper, I take a similar approach and estimate individual-level policy rules using SEP forecasts.¹⁰

The Zero Lower Bound (ZLB) imposes a natural censoring constraint on the policy rate. Short-term forecasts of the federal funds rate were near the zero level throughout more than a third of the sample, as can be seen in Figure 2. To address this constraint,

⁹However, individual forecasts of other variables are available before 2012.

¹⁰To my knowledge, this is the first paper to estimate and analyze individual policy reaction functions using FOMC forecast data. Fendel and Rülke (2012), Arai (2023), and González-Astudillo and Tanvir (2024) use similar data (Summary of Economic Projections, or the Monetary Policy Reports), but estimate a single reaction function for the specified group of members, and therefore lack any insights on heterogeneity of reaction functions inside Fed.

I estimate individual policy rules using a left-censored Tobit model — a standard approach for modeling monetary policy during periods when the policy rate is constrained by the ZLB (e.g., Huang 2015, Kim and Pruitt 2017, Mavroeidis 2021, Arai 2023).¹¹ Ultimately, the Tobit model is preferred because it provides consistent estimates by properly handling the censored nature of the dependent variable.

I estimate the following Tobit model:

$$\widehat{FFR}_{it}^h = \begin{cases} \widetilde{FFR}_{it}^h, & \text{if } \widetilde{FFR}_{it}^h > \tau_{\text{lower}} \\ \tau_{\text{lower}}, & \text{if } \widetilde{FFR}_{it}^h \leq \tau_{\text{lower}} \end{cases}, \quad (1)$$

where the latent policy rate forecast is given by:

$$\widetilde{FFR}_{it}^h = X_{it}^h \beta_i + \epsilon_{it}^h \equiv \alpha_i + \alpha_t + \gamma_i \cdot (\widehat{UN}_{it}^h - \widehat{UN}_{it}^{LR}) + \omega_i \cdot (\widehat{\pi}_{it}^h - \widehat{\pi}_{it}^{LR}) + Z_{it} \delta + \epsilon_{it}^h. \quad (2)$$

Throughout, i indexes FOMC members, t indexes SEP release date, and the superscript h denotes the forecast target year. I consider horizons $h \in \{0, 1, 2, 3\}$ for current, next, two and three years ahead forecasts. The latent policy rate forecast is modeled as a linear function of the inflation and unemployment gaps, reflecting the Federal Reserve’s dual mandate. The observed federal funds rate forecast ($\widehat{FFR}_{it}^h$) equals the latent forecast ($\widetilde{FFR}_{it}^h$) if it is greater than the censoring threshold (τ_{lower}) imposed by the ZLB, and equals the threshold otherwise. I use the long-term forecasts of each policymaker to proxy their target values. This allows us to utilize their beliefs about targets and avoid imposing fixed target values, which might not suit all policymakers. The coefficients γ_i and ω_i represent the weights on unemployment and inflation gaps, respectively, in the latent policy rule and are allowed to vary across FOMC members. Z_{it} includes a voting status dummy and its interactions with the forecasted inflation and unemployment gaps, allowing the responsiveness of forecasts to vary by voting status. The error term is assumed to be i.i.d. Normal, $\epsilon_{it}^h \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma^2)$.

Time fixed effects α_t and member fixed effects are also included in the model, allowing the natural rate of interest to vary across both time and individuals. This contrasts with the standard formulation of the Taylor rule, which assumes a constant natural rate, see Taylor (1993). My approach is motivated by a growing body of evidence suggesting that the natural rate, while relatively stable in the short run, is subject

¹¹The Tobit model accounts for censoring in the dependent variable. While OLS estimates differ somewhat, with the unemployment gap coefficient seems to be biased toward zero for most of the FOMC members, the qualitative conclusions of the paper remain robust, see Figure 1 in the Online Appendix.

to substantial fluctuations over time, see Holston, Laubach, and Williams 2017. This variation is also evident in the SEP forecasts of the long-run interest rate within my sample, see Figure 4. These projections can be interpreted as forecasts of the natural rate, under the assumption that inflation and unemployment gaps close in the long run. The data show that projected long-run interest rates vary significantly over time and, to some extent, across members, with the median forecast ranging from a minimum of 2.7% to a maximum of 4.25%. Assuming a constant natural rate can lead to biased estimates of the coefficients of inflation and economic activity, Trehan and Wu (2007). To address this concern, I exploit the panel structure of the data by incorporating time fixed effects, which control for shifts in the natural rate and/or policy persistence.

The contemporaneous formulation of the policy rule is chosen because it preserves all available observations while yielding a similar pseudo R^2 to alternative forward-looking or recursive specifications.¹² Conceptually, this formulation treats forecasted inflation and real economic activity as exogenous, outside the central bank’s control, reflecting the lagged effects of monetary policy. In contrast, using the current or short-run interest rate alongside long- or medium-run forecasts of inflation and unemployment likely introduces endogeneity, Shapiro and Wilson (2022). I treat inflation and unemployment expectations as predetermined inputs to the policy rule, consistent with the SEP data structure. Modeling expectation formation is beyond the scope of this paper.

The model is estimated on the sample of both voting and non-voting members. One potential concern here is that there might be systematic differences in individual forecasts submitted by a member whose voting status changes over time. For example, Tillmann (2011) found that non-voters systematically overpredict inflation relative to the consensus forecast if they favor tighter policy and underpredict inflation if they favor looser policy. Forecasts from voting members, in contrast, show no systematic relationship with policy preferences. This is evidence on strategic forecasting of FOMC members. To account for potential strategic forecasting, I include Z_{it} — a set of interactions between voting status and the explanatory variables.¹³

¹²The structure of the SEP data complicates estimation of forward-looking policy rules, as forecasts are made in different quarters but always target the fourth quarter values.

¹³I formally test whether the deviations in forecasts from the consensus depend on the voting status in what follows, and find no evidence that voting status systematically affects the forecasts of regional Federal Reserve Bank presidents. This is further confirmed by the fact that all effects of interaction terms in Z_{it} in the latent policy rate equation are statistically insignificant and economically small.

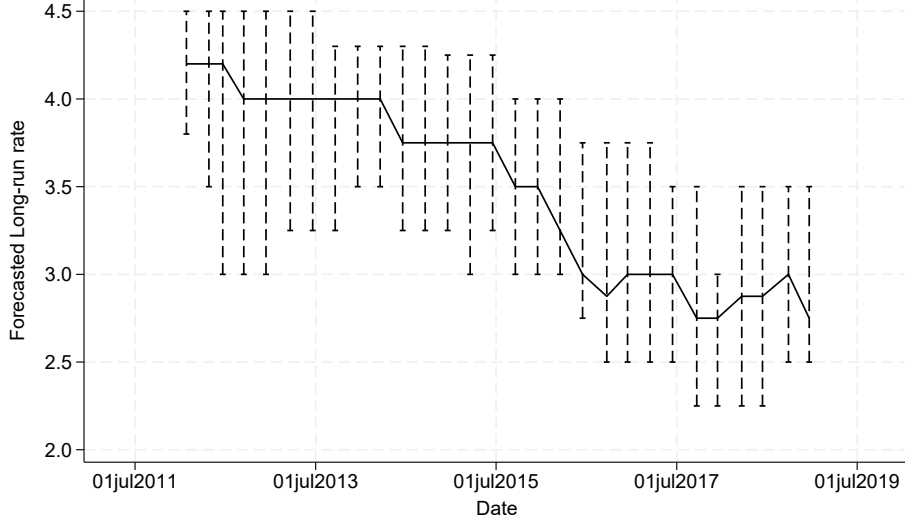


Figure 4: Long-run interest rate projections over time

Notes: Dashed intervals indicate the range between the minimum and maximum forecasts.

The model is estimated using the Maximum Likelihood Estimator. Because the regression errors may be correlated across horizons within a member–date cell (i, t) , we compute standard errors clustered at the member $\times$ date level, which allows for arbitrary correlation across horizons within each cell.¹⁴ First, define the indicator function:

$$\mathcal{I}(\widehat{FFR}_{it}^h) = \begin{cases} 1, & \text{if } \widehat{FFR}_{it}^h > \tau_{\text{lower}} \\ 0, & \text{if } \widehat{FFR}_{it}^h = \tau_{\text{lower}}. \end{cases} \quad (3)$$

The likelihood function is given by:

$$\mathcal{L} = \prod_{i=1}^N \prod_{t=1}^T \prod_{h=1}^H \left(1 - \Phi \left[\frac{X_{it}^h \beta_i - \tau_{\text{lower}}}{\sigma} \right] \right)^{1 - \mathcal{I}(\widehat{FFR}_{it}^h)} \cdot \left(\frac{1}{\sigma} \varphi \left[\frac{\widehat{FFR}_{it}^h - X_{it}^h \beta_i}{\sigma} \right] \right)^{\mathcal{I}(\widehat{FFR}_{it}^h)}, \quad (4)$$

where $\Phi(\cdot)$ and $\varphi(\cdot)$ denote the standard normal cumulative distribution function and probability density function, respectively. τ_{lower} is the censoring threshold, which is treated as fixed and exogenous. In the data, the federal funds rate forecasts are measured as the midpoint of the target range. For example, if the forecasted range is

¹⁴Almost identical confidence intervals are obtained when clustering at the meeting level. A more serious concern that could potentially lead to inconsistent estimates is heteroskedasticity. However, the standard errors with and without heteroskedasticity-robust adjustments (Eicker–Huber–White) are very similar (see Figure 6 in Online Appendix A), suggesting that the error terms are close to homoscedastic.

$[0, 0.25]$, the reported value of the forecast is 0.13 — the midpoint of the $[0, 0.25]$ range, rounded to two decimal places. Accordingly, I set $\tau_{\text{lower}} = 0.13$.

3.2 Average marginal effects

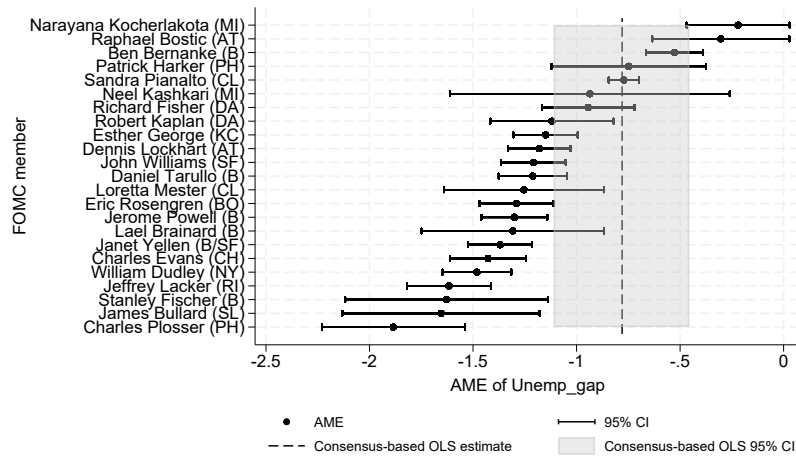
I focus on the average marginal effects of the forecasted federal funds rate with respect to a unit change in either the unemployment or inflation gap. This captures how each member adjusts their policy rate forecast in response to deviations in these key inputs. Figure 5 shows estimated inflation and unemployment gap AMEs across FOMC members. As a benchmark, the dashed lines show AMEs from a pooled linear regression estimated with consensus (mean) SEP forecasts. First, for the sample of 23 FOMC members, all but one of the unemployment gap coefficients are negative, and all inflation gap coefficients are positive. This aligns well with the theoretical predictions, and mean AMEs are consistent with recent empirical results (on aggregated data) from the literature, see for example Carvalho, Nechio, and Tristao (2021).

Second, the magnitude of the estimated coefficients varies substantially across members. The unemployment gap coefficient ranges from -0.3 to -1.9, with an average of -1.2. This implies that a 1 percentage point increase in the unemployment gap is associated with a reduction in the forecasted federal funds rate of between 0.3 and 1.9 percentage points, depending on the member, suggesting substantial variation in the perceived costs of labor market slack. For the inflation gap, the effect ranges from 0 to 3.4 percentage points, with an average of 2.1.

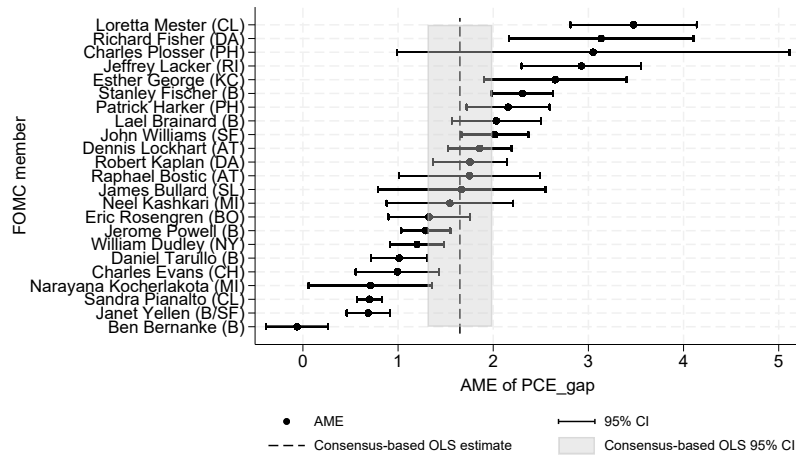
Partly, this heterogeneity could be attributed to a limited sample size - it is hard to estimate coefficients precisely if the number of observations per member is low. However, with only a few exceptions, all coefficients are statistically significant, with many coefficients also statistically different from each other. Moreover, the alternative regression with fixed weights provides a significantly worse fit to the data.¹⁵ The goodness-of-fit measures are statistically higher for a more flexible specification.¹⁶ Finally, this heterogeneity is not driven by one or two members with extreme values of marginal effects. Taken together, these findings suggest meaningful intrinsic differences across FOMC members—including those within the same institutional structure, such as the Board or a specific regional Federal Reserve Bank.

¹⁵The fixed weights Tobit model produces AMEs of -1.37 and 2.12 for the unemployment gap and inflation, respectively.

¹⁶Based on the likelihood ratio test and the Akaike Information Criterion, the model with individual coefficients fits the data significantly better than the simpler model with fixed coefficients for all members. For example, AIC is 30% lower for the more flexible specification, indicating superior performance even after penalizing for additional parameters. Both the LR test and the AIC criterion suggest that this more flexible specification is preferable over the fixed weights model specification. This means that the cross-sectional heterogeneity of AMEs is statistically large in the data.



(a) Unemployment gap AMEs



(b) Inflation gap AMEs

Figure 5: Individual policy rule AMEs

To test the robustness of these results, I perform several exercises. First, I change the threshold on the number of observations required per FOMC member (I use 35, 40, 45, 50 values); this has a limited effect on the results. Second, I estimate the policy rules with OLS instead of the Tobit model; see Figure 1 in the Online Appendix A for the comparison. Estimated coefficients are not substantially different across the two estimation methods, and I still observe substantial heterogeneity in the estimates. Third, policymakers can form their forecasts of the federal funds rate based on the core inflation rate, which is arguably better suited for the purpose of analyzing the inflation trends. Substituting the PCE inflation gap with core PCE inflation yields nearly identical results; see Figure 2 in the Online Appendix A. Using $\pi^* = 2$ inflation target as stated in the Statement on Longer-Run Goals and Strategy¹⁷ instead of long-

¹⁷https://www.federalreserve.gov/monetarypolicy/files/fomc_longerrungoals.pdf

run inflation forecasts also produces similar results¹⁸.

Next, I also estimate a version of the policy rules using GDP gap forecasts in place of unemployment gaps. The results differ markedly: for many members, the estimated coefficients on the GDP gap are negative; see Figure 3 in the Online Appendix A. This would imply that output above potential is associated with a looser policy stance, which contradicts standard monetary policy intuition. At the same time, the inflation gap coefficients become much larger. Similar patterns are documented in Arai (2023). One likely explanation is that many FOMC members substantially revised down their long-run output growth projections after 2012, when dot plot projections began. Because the output gap is constructed as the difference between short- and long-run projections, downward revisions to potential output mechanically increased the estimated gap, potentially reversing its sign. Importantly, the Tobit model with the unemployment gap fits the data better than the model with the GDP gap: the pseudo R^2 is 0.6381 versus 0.5197, and the AIC is 2172 versus 2818, respectively.

Another potential concern in estimating a Tobit model with member-specific intercepts and slopes is the incidental parameters problem, Greene (2002). The incidental parameters problem arises when the number of parameters to be estimated increases with the number of individuals, potentially leading to biased and inconsistent estimates, especially in short panels with limited observations per unit. In our case, three parameters are estimated per member: a fixed effect, the unemployment gap slope, and the inflation gap slope. However, I also require each FOMC member to contribute at least 40 forecasts, with the average number of forecasts per member being around 48 over the sample. This relatively large number of observations per member mitigates the severity of the incidental parameters problem. As a result, while some finite-sample bias may remain, it is unlikely to materially affect the estimated coefficients or the overall conclusions.

Finally, because SEP forecasts are formed simultaneously, there is a potential for reverse causality: changes in expected macro variables affect the expected policy rate, and changes in the expected policy rate might alter the projections of macro variables. However, the contemporaneous specification alleviates this concern, as it is unlikely that changes in the policy rate forecast would affect same-period projections of macroeconomic conditions. Also, there is evidence that the endogeneity in various forms of policy rules, estimated on the aggregate data, is likely to be limited; see Carvalho, Nechio, and Tristao (2021). To reduce the effect of this potential simultaneity, I provide results where the forecasted inflation and unemployment gaps are replaced with their

¹⁸Most members forecast around 2% inflation rate in the long run in all time periods.

first lags. The resulting AMEs are quantitatively similar to the baseline, confirming the robustness of our estimates, see Figure 4 in the Online Appendix A.

3.3 Implications of Committee Structure for Forecasted Policy Rates

In this part, I explore how heterogeneity in individual reaction functions translates into differences in the forecasted policy rate. I evaluate this by considering several counterfactual short-term predictions of the federal funds rate from the Tobit model, estimated previously. Figure 6 shows the following predictions: (1) average federal funds rate prediction from the estimated Tobit model, (2) counterfactual predicted federal funds rate if I use Chair’s AMEs for all members in the committee, (3) counterfactual prediction if committee consists of only a Chair, (4) actual consensus federal funds rate forecast.

To construct counterfactual (2) I use the previously estimated Tobit model and generate predicted values of the fed rate such that I use forecasts of all FOMC members in the period, but restrict AMEs of these members to be exactly equal to the Chair’s estimated AMEs. Counterfactual (3) uses the estimated equation corresponding to the Chair. In other words, I ignore all other members’ predictions and estimated AMEs and assume that the committee consists of only the Chair, and calculate the predicted values for this scenario.

The two counterfactual scenarios imply different forecasts, compared both with actual consensus forecasts (4) and with the average predicted forecast from the Tobit model (1). In some time periods, the difference exceeds 75 basis points. All this indicates that heterogeneity in reaction functions results in different forecasts of the policy rate.

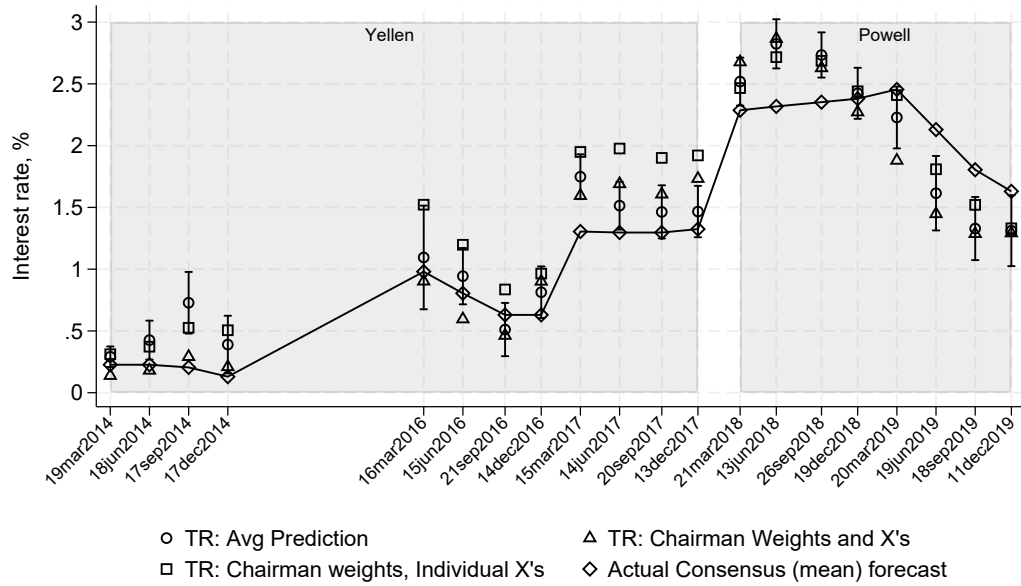


Figure 6: Counterfactual Forecasts of the Federal Funds Rate.

Notes: Actual/counterfactual forecasts made in year T for the fourth quarter of year T

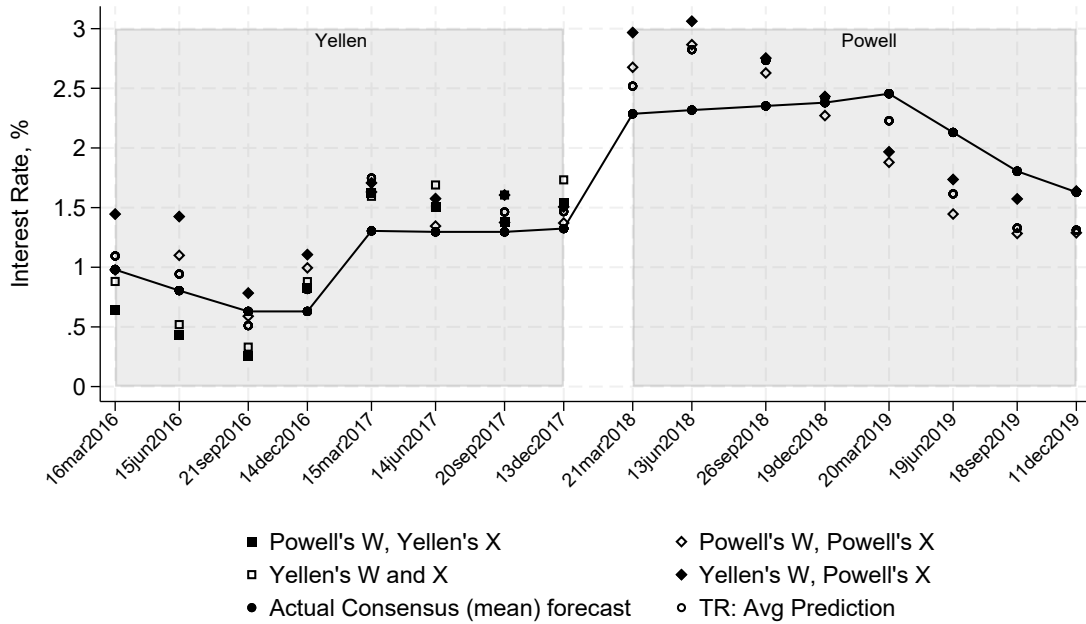


Figure 7: Counterfactual Forecasts of the Federal Funds Rate.

Notes: Actual/counterfactual forecasts made in year T for the fourth quarter of year T . W = reaction-function weights; X = input forecasts (inflation & unemployment gaps).

Figure 7 focuses on the period 2016–2019, when two individuals served as Chair: Janet Yellen and Jerome Powell. In addition to baseline Chair-only scenarios, I intro-

duce two additional counterfactuals that “swap” reaction functions across Chairs while holding their macroeconomic projections fixed. In the “Powell’s W, Yellen’s X” case, I apply estimated Powell’s AMEs to Yellen’s forecasts; in the reverse case, Yellen’s AMEs are applied to Powell’s forecasts. These swaps yield non-trivial differences in projected interest rates—reinforcing the point that differences in reaction functions, not just macro outlooks, may drive policy variation. For instance, if J. Powell “uses” J. Yellen’s reaction function, while keeping his views on the economy intact, the change in the forecasted policy rate by him would be on average 25-50 basis points for most of the sample.

4 Summary measure of reaction functions heterogeneity

As established in Section 3.2, average marginal effects vary substantially across members. There is clear variation across members in both input dimensions, see Figure 8. Some members, such as Charles Plosser and Richard Fisher, display strong reactions to inflation deviations, while others, including Ben Bernanke and Narayana Kocherlakota, exhibit relatively muted responses to both inflation and unemployment. Members like Janet Yellen, William Dudley, and John Williams exhibit more moderate and balanced policy reaction functions.

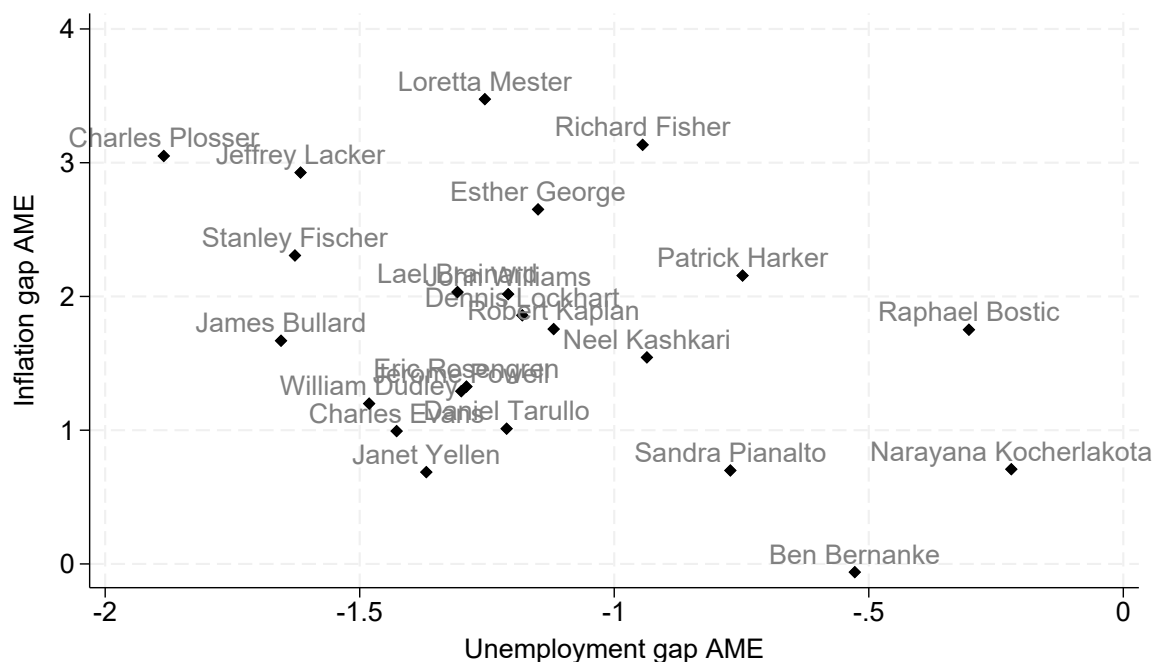


Figure 8: Scatterplot of average marginal effects.

Variation in marginal effects across members captures differences in the absolute importance of each input—it does not reflect the inflation-unemployment tradeoff per se. For example, it is not clear whether Narayana Kocherlakota is more hawkish or dovish than Neel Kashkari. The former has smaller responses to both objectives than the latter. To summarize each member’s policy reaction functions along a single dimension, I construct a continuous, parsimonious measure of the perceived tradeoff between inflation and unemployment based on the estimated policy rules.

There are several ways to construct such a measure. A common approach defines a member as hawkish if they place greater emphasis on controlling inflation, and as dovish if they prioritize reducing unemployment or supporting economic activity.¹⁹ Given estimated individual policy rules, a natural way to define such a measure is to take a sum of AMEs of inflation and unemployment gaps.²⁰ This measure is depicted in Figure 9.

I find that the relative importance of inflation vs unemployment differs across members, stemming from substantial variation in reaction functions. Even across the Board members or across the members from the same regional Fed our measure sometimes implies statistically significantly different inflation/unemployment tradeoff. One notable pattern is that Board members tend to be more dovish than regional Fed presidents, consistent with past qualitative classifications, Bordo and Istrefi (2023).

It is important to emphasize that our constructed Hawk measure reflects perceived (or revealed) tradeoff and not necessarily the true preferences of FOMC members, which are not observable. The constructed *Hawk* measure is thus viewed as relative valuation of inflation versus unemployment stabilization, *regardless of the underlying reasons for this perceived tradeoff*. Members who place greater weight on controlling inflation for whatever reasons, relative to reducing unemployment, are considered more hawkish, while those who prioritize minimizing unemployment are considered to be more dovish.

¹⁹This characterization is regardless of members’ views on the effects of monetary policy.

²⁰While a simple sum of AMEs captures overall policy stance, it might not capture differences in magnitudes. For example, if a member’s A and member’s B sum of AME is approximately equal (like in the case of Narayana Kocherlakota and Neel Kashkari), it might not mean, in general, that these two members react equally to the same change in a policy rule input. One input might be emphasized more strongly by one member than the other, i.e., one member can have a higher AME for this input. One way to take that into account is to standardize estimated AMEs:

$$Hawk'_i = (AME_{UN,i} - mean(AME_{UN,i}))/SD(AME_{UN,i}) + (AME_{\pi,i} - mean(AME_{\pi,i}))/SD(AME_{\pi,i}).$$

This alternative measure is shown in Figure 5 in Online Appendix A and highly correlated with the original one (corr>0.85). All main conclusions remain the same regardless of the choice of measure.

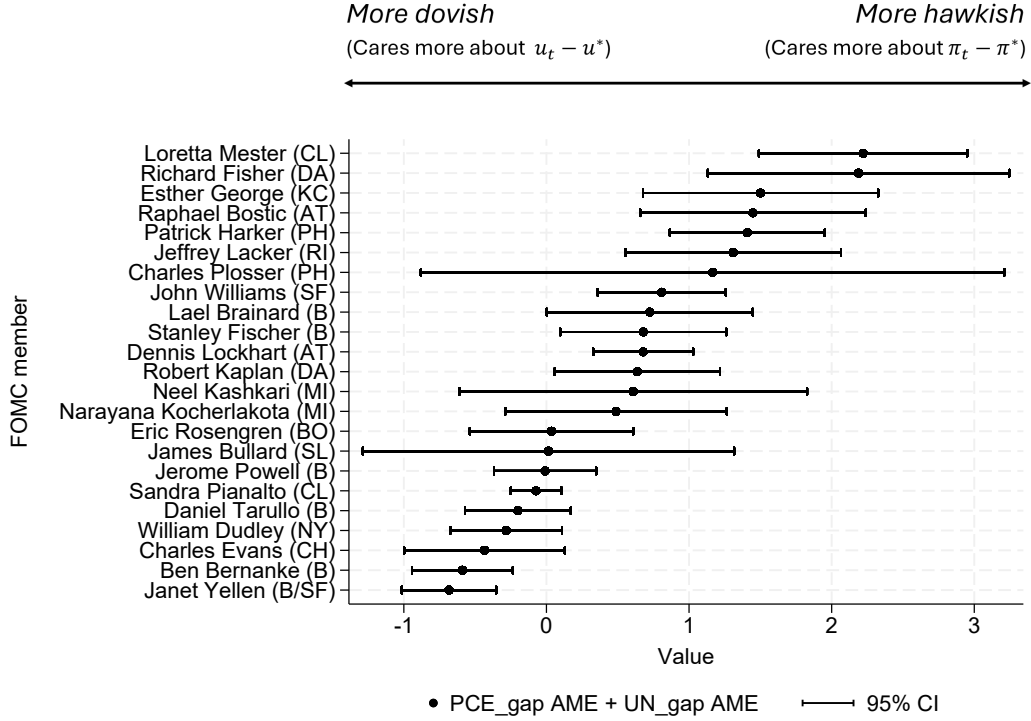


Figure 9: Estimated preferences measure

For example, the level of this measure for a particular member is likely affected by what the member thinks about other members' decisions. Consequently, it cannot be directly viewed as a measure of true preferences. For example, FOMC members do not vote in isolation — they understand that the final policy outcome depends on committee consensus. Therefore, a member's forecasted rate (dependent variable) may reflect not only their own preference, but also anticipation of the median or majority vote, and/or strategic positioning to influence or signal to others. Thus, heterogeneity in estimated reaction functions could reflect differences in expectations about other members, rather than differences in preferences per se. In addition to strategic considerations, differences in estimated functions may also stem from informational asymmetry. Policymakers might have access to different sets of information. For example, though all have access to the same staff forecasts, each FOMC member has their own set of advisors, which can result in different estimated reaction functions.

Next, FOMC members operate in distinct macroeconomic environments—for example, periods of low inflation and low unemployment, or high inflation and low unemployment. This can mechanically lead to differences in estimated reaction functions, even if the underlying preferences over inflation and unemployment are perceived similarly across members. For example, it is not clear whether a member places low weight on inflation because inflation was near target during their tenure, or because of a gen-

une preference to deprioritize it. For instance, Bernanke exhibits a near-zero inflation response (AME of inflation), but it is unclear whether this reflects his intrinsic preferences or the characteristics of the sample period (low inflation) in which he served. Consequently, observed heterogeneity in reaction functions may partly reflect variation in the economic environments faced by policymakers.²¹ That said, the sample period analyzed does not include major crisis episodes (starting after the end of the GFC, and ending before Covid-19) and is relatively homogeneous. This should limit the extent to which environmental factors drive differences in reaction functions.

On the other hand, federal funds rate forecasts can also represent what the FOMC member *would like* the interest rate to be. The Fed provides the following interpretation of SEP forecasts: “[Interest rate forecasts are] based on policymakers’ assessments of appropriate monetary policy, which, by definition, is the future path of policy that each participant deems most likely to foster outcomes for economic activity and inflation that best satisfy his or her interpretation of the Federal Reserve’s dual objectives of maximum employment and stable prices”. This supports the interpretation that forecasts represent the views on appropriate policy and preferred interest rates of FOMC members. Similarly, Bernanke (2016) argues that SEP provides indirect information about the FOMC reaction’s function.

However, in general, it is not clear how exactly FOMC members form their forecasts. For example, from Blinder (2022): “Notably, in January 2012 it began offering market participants its now-famous ‘dot plot’, showing where FOMC members **expected (or was it wanted?)**²² the funds rate to be over the next few years.” Thus, an important question is whether these estimated preferences accurately reflect the underlying policy inclinations of FOMC members²³.

I argue that the differences in estimated reaction functions might indeed suggest differences in preferences of FOMC members. I provide several empirical pieces of evidence supporting this view. First, as I discuss in the following Subsection 4.1, individual characteristics help explain a substantial part of the heterogeneity in reaction functions. Fundamental and exogenous individual characteristics (such as education and working experience) are indeed known determinants of preferences (Bordo and Istrefi 2023, Malmendier, Nagel, and Yan 2021). Individual characteristics would be mostly irrelevant

²¹If that is indeed the case, it would suggest that preferences are not structural but instead state-dependent, implying that policymakers’ responses vary non-linearly with the economic environment. Modeling such non-linearities would require a more complex specification of the policy reaction function, which is difficult to estimate given the limited data available.

²²Emphasis added by the author to highlight the interpretive ambiguity.

²³Over the last couple of years, [there has been a discussion](#) on how to improve SEP design, including, but not limited to making it better suited to inform of members reaction functions.

if members primarily cared only about other members’ views or if estimated AMEs were affected by the economic environment. Second, in Subsection 4.2 I show that more hawkish FOMC members, according to our measure, perceive more inflation risk, which motivates a higher policy rate—i.e., they have hawkish expectations. Third, in Subsection 4.3, I also find that voting decisions are associated with the reaction function implied inflation-unemployment tradeoff measure. Finally, in Subsection 4.4 I show that evidence of strategic forecasting is limited in the data and unlikely to be a primary driver of observed differences.

How does our measure correspond with the literature? One prominent example is Istrefi (2019), which characterizes members as hawks, doves, and swingers based on textual analysis of more than 20,000 Fed documents.²⁴ Although the full classification is not publicly available, several similarities can be observed. First, I find that board members are clustered on the dovish side of the spectrum, which coincides with the findings Bordo and Istrefi (2023) that board members are likely to be doves. Second, as we will see later, more hawkish members tend to dissent in favor of more restrictive policy more often, while also providing higher inflation forecasts. Finally, our measure echoes findings from Bordo and Istrefi (2023), including the importance of educational background in shaping preferences.

4.1 Explaining the Heterogeneity in Estimated AMEs

Heterogeneity in the reaction functions among Federal Reserve (Fed) members likely reflects a range of structural, institutional, and personal influences. Prior research points to differences in academic training, economic philosophy (Bordo and Istrefi 2023, Chappell Jr., McGregor, and Vermilyea 2008), regional economic conditions (Bobrov, Kamdar, and Ulate 2024), career incentives (Abrams and Iossifov 2006), and political pressures (Schnakenberg, Turner, and Uribe-McGuire 2017) as key contributors to preference heterogeneity among FOMC members.

Lifetime experiences and memories play an important role in determining the formation of FOMC preferences. For instance, Malmendier, Nagel, and Yan (2021) find that past experiences of high inflation make FOMC members more hawkish, with the effect diminishing as time passes since the event. Bordo and Istrefi (2023) point out that not only inflation but also unemployment experiences are shaping the views of committee members. To reflect these ideas, I introduce two variables: the member’s age and the number of years the member has spent in the Federal Reserve (*Tenure*).

²⁴This classification is mainly based on media coverage, thus reflecting their perceptions of the policymakers’ intentions.

As shown in Bordo and Istrefi (2023), the preferences of FOMC members are most likely formed during their university studies; the education they obtain may be an important determinant of their professional views and policy choices. To capture that, I use the type of university degree (PhD in economics or related fields, JD degree, other degrees), as well as the “freshwater”/“saltwater” type distinction. I draw on the classification of economic ideology based on educational background, specifically referencing the “freshwater” and “saltwater” schools of thought, as discussed in Bordo and Istrefi (2023). This terminology reflects the geographical distribution of universities with differing macroeconomic perspectives. “Freshwater” institutions, primarily located near the Great Lakes in the United States, are contrasted with “saltwater” universities, which are situated closer to the coasts. This distinction is naturally associated with the prevalence of monetarist or Keynesian schools of thought.

In this part, I assess the importance of individual characteristics of FOMC members commonly used in the literature to explain observed heterogeneity in AMEs of policy rules across Fed members. I consider the following characteristics: Age, Tenure, Political affiliation, Gender, PhD/JD/Other university degree, “Freshwater” vs “Saltwater” affiliation, Board vs Regional affiliation, and economic research experience. Descriptive statistics of the members are shown in Table 1 in the Online Appendix B.

Of course, these characteristics do not fully capture the nuances of each member’s preferences. For instance, two members may value unemployment stabilization equally but differ in their beliefs about the effectiveness of monetary policy in achieving it. Such differences—rooted in views about transmission mechanisms—are not directly observable and are difficult to proxy without a separate structural model. However, the characteristics used in this paper mainly come from the literature, and evidently, most of them correlate with the preferences of FOMC members.

Given these constraints in mind, I estimate the following non-linear model to test whether individual characteristics can explain heterogeneity in estimated individual policy rule AMEs:

$$\begin{aligned} \widetilde{FFR}_{it}^h = & \alpha_i + \alpha_t + (\gamma_1 Age + \gamma_2 PhD + \dots) \cdot (\widehat{UN}_{it}^h - \widehat{UN}_{it}^{LR}) + \\ & (\omega_1 Age + \omega_2 PhD + \dots) \cdot (\widehat{\pi}_{it}^h - \widehat{\pi}_{it}^{LR}) + Z_{it}\delta + \epsilon_{it}^h. \end{aligned} \quad (5)$$

I follow the approach in Section 3 and estimate a left-censored Tobit model. Average marginal effects (AMEs) for this specification are reported in Online Appendix C—Table 2 for unemployment-gap interactions and Table 3 for inflation-gap interactions. In columns (1)–(8), I include one determinant at a time; the final column includes all individual characteristics jointly. Many characteristics are statistically sig-

nificant when considered individually, and several interaction terms remain significant after controlling for all characteristics simultaneously. In the full-controls regression, the most influential factors are educational background, research experience, institutional position at the Federal Reserve, and work experience. For example, PhD holders in economics or related fields place substantially less weight on inflation and more weight on economic activity. Older and more experienced FOMC members—many of whom lived through major downturns such as the 1970s (Malmendier, Nagel, and Yan 2021)—tend to put greater emphasis on economic activity. Board members place relatively less weight on inflation, whereas research experience is associated with placing more weight on it.

By contrast, place of graduation is not associated with the relative weights on the two objectives once controls are included, which differs from Bordo and Istrefi (2023). While the “freshwater” versus “saltwater” distinction may have been more informative in earlier decades (their sample includes members served in the 1970s–1990s), the divergence in results here likely reflects the inclusion of both member and time fixed effects. Indeed, a simple regression of the *Hawk* measure on freshwater/saltwater indicators reproduces the Bordo and Istrefi (2023) pattern—freshwater graduates are, on average, more hawkish than saltwater graduates. However, the effect vanishes once I control for fixed characteristics. For instance, none of the Board members in my sample graduated from a “freshwater” university, so without controls, the estimated “school-of-thought” effect can be conflated with institutional position.

To evaluate how well observable characteristics account for heterogeneity in reaction functions, I compare the AMEs from the flexible, member-specific specification with those implied by the model above. Figures 11 and 10 show that these two sets of AMEs are positively correlated, indicating that much of the observed variation in average marginal effects can be attributed to a small number of observable member characteristics. This suggests that differences in reaction functions could indicate differences in the preferences of FOMC members.

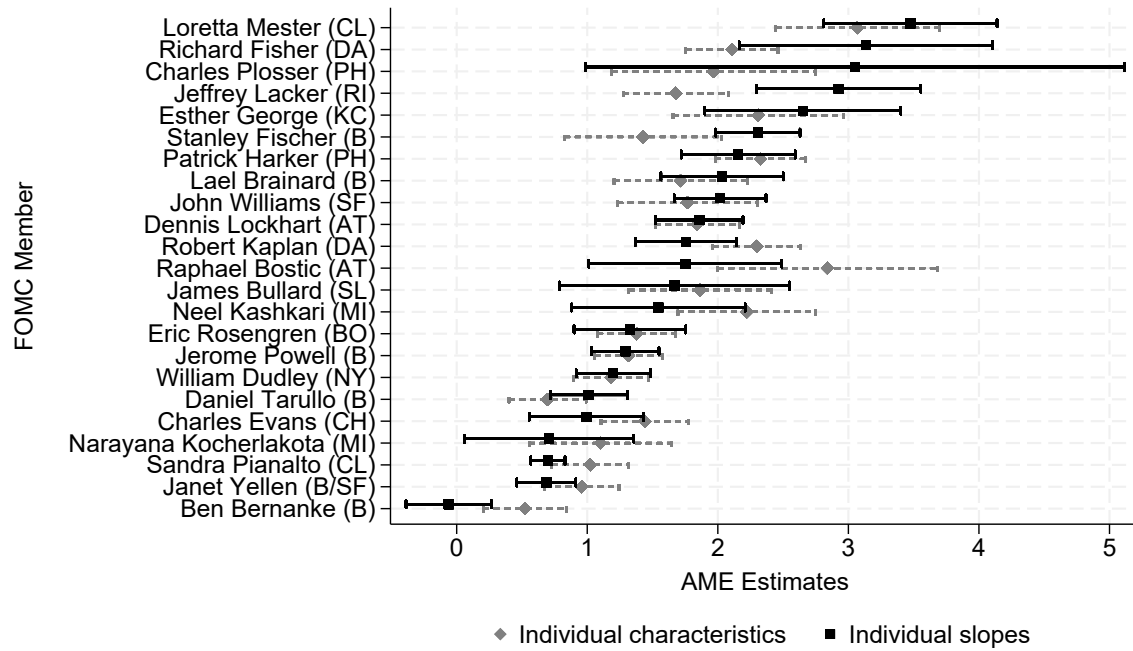


Figure 10: Inflation gap coefficients: individual slope vs individual characteristics.

Notes: confidence intervals correspond to 95% level

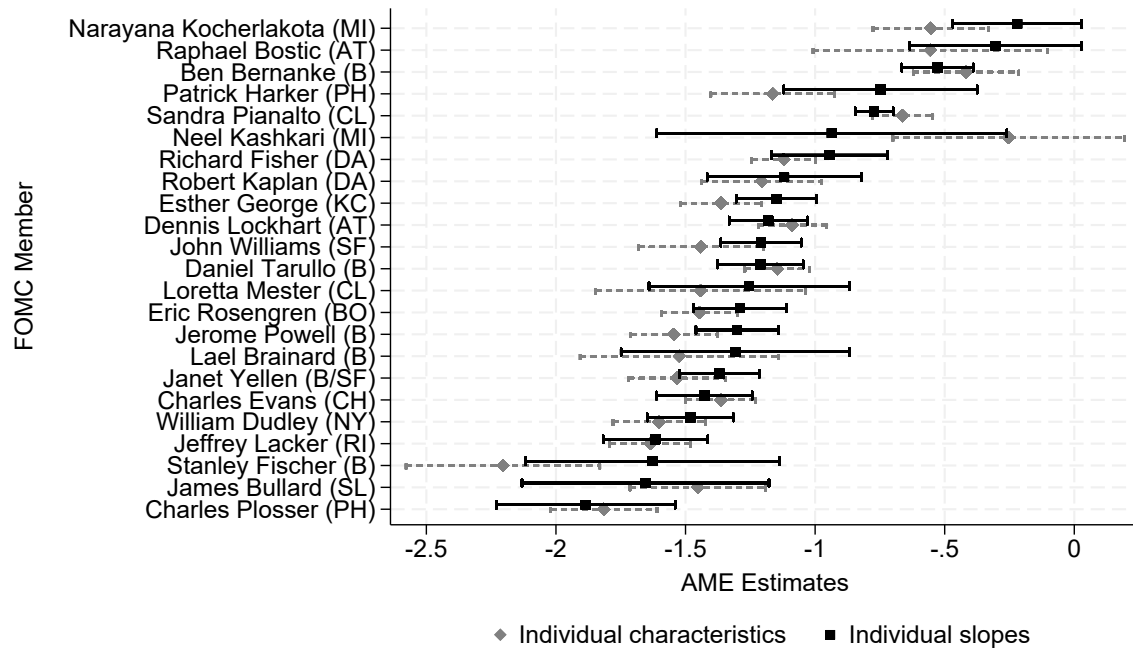


Figure 11: Inflation gap coefficients: individual slope vs individual characteristics.

Notes: confidence intervals correspond to 95% level

4.2 FOMC forecasts and *Hawk* measure

In the following, I test whether forecast differences across FOMC members are systematically related to their position on the estimated preference spectrum, as captured by the *Hawk* measure. Table 1 shows the OLS estimates of the regression of our estimated preference measure, *Hawk*, on i) inflation, ii) unemployment, and iii) GDP growth forecasts. I control for common to all members characteristics of the economic environment in time t by including time fixed effects.

These results provide some support for the commonly held view that hawkish policymakers are more likely to anticipate inflationary pressures. An increase in our *Hawk* measure is associated with higher forecasted inflation. The effects are statistically significant (p-value < 0.01), though small in economic terms.

Table 1: FOMC forecasts and *Hawk*

	Inflation	Unemployment	GDP
Current Year Forecasts			
β estimate	0.054***	-0.022***	-0.015
SE	(0.010)	(0.007)	(0.011)
Next Year Forecasts			
β estimate	0.092***	-0.012	-0.066***
SE	(0.013)	(0.012)	(0.018)
Time FEs	Yes	Yes	Yes

Robust standard errors in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. N = 637.

4.3 Dissent voting and *Hawk* measure

Dissent data from the Federal Reserve systematically records instances in which individual members of the Federal Open Market Committee (FOMC) express disagreement with the prevailing consensus on monetary policy decisions. This information is captured through formal statements and detailed voting records during FOMC meetings. Analyzing dissent data provides critical insights into the heterogeneity of perspectives within the committee and elucidates the underlying factors that influence policy outcomes.

However, preferences are not the only factor explaining why certain members deviate from the consensus. Other influences, such as strategic considerations, career incentives, institutional roles, or misinterpretations of economic conditions, can also shape dissenting behavior. Dissenting votes are typically deliberate and reflect deeper policy

disagreements. Sometimes they serve a strategic purpose, allowing members to signal their hawkish or dovish leanings, shape future discussions, or build their reputations within the institution. Another reason for dissent might be alternative interpretations of economic conditions. In some cases, dissenting members may even be ahead of the consensus, identifying risks or opportunities that others overlook.

In order to demonstrate the role of heterogeneity of reaction functions, I show how the estimated tradeoff measure (*Hawk*) is related to the voting behavior of FOMC members. To do that, I estimate the ordered probit model, where the dependent variable is $Dissent \in \{-1, 0, +1\}$ (-1 = dovish dissent, 0 = consensus, +1 = hawkish dissent). The independent variable is the estimated tradeoff measure (*Hawk*), which is the sum of inflation and unemployment gap AMEs. Formally, I assume an ordered-probit structure with latent index:

$$z_{it}^* = X_{it}\eta + \eta_H \text{Hawk}_i + \mu_t + \mu_i + \nu_{it}, \quad \nu_{it} \sim \mathcal{N}(0, 1),$$

where X_{it} collects individual characteristics used earlier in equation (5) (education, age, working experience, etc.), and μ_t are meeting fixed effects capturing common shocks.

Observed outcomes follow threshold crossings:

$$\text{Dissent}_{it} = \begin{cases} -1, & \text{if } z_{it}^* \leq \kappa_1, \\ 0, & \text{if } \kappa_1 < z_{it}^* \leq \kappa_2, \\ +1, & \text{if } z_{it}^* > \kappa_2, \end{cases} \quad \kappa_1 < \kappa_2,$$

with the usual ordered-probit normalization $\text{Var}(\nu_{it}) = 1$. The implied probabilities are:

$$\begin{aligned} \Pr(-1 \mid \cdot) &= \Phi(\kappa_1 - \eta' X_{it} - \eta_H \text{Hawk}_i - \mu_t - \mu_i), \\ \Pr(0 \mid \cdot) &= \Phi(\kappa_2 - \eta' X_{it} - \eta_H \text{Hawk}_i - \mu_t - \mu_i) - \Phi(\kappa_1 - \eta' X_{it} - \eta_H \text{Hawk}_i - \mu_t), \\ \Pr(+1 \mid \cdot) &= 1 - \Phi(\kappa_2 - \eta' X_{it} - \eta_H \text{Hawk}_i - \mu_t - \mu_i), \end{aligned} \quad (6)$$

where $\Phi(\cdot)$ is the standard normal CDF.

The first column of Table 2 shows the marginal effects of the *Hawk* measure, with individual characteristics as control variables. The second column adds time fixed effects. I find that the *Hawk* measure affects the probability of dissent in both specifications. For instance, in the second specification, a one-unit increase in the *Hawk* measure (approximately 0.8 standard deviations) is associated with a statistically significant in-

crease in the probability of a hawkish dissent (+3.8 percentage points) and a decrease in the probability of a dovish dissent (−3.2 percentage points).

Table 2: Average marginal effects of *Hawk* measure on the dissent types

	<i>Hawk AMEs</i>	
Dovish Dissent	-2.5 ***	-3.2 ***
Consent	-1.0	-0.6
Hawkish Dissent	3.5 ***	3.8 ***
Pseudo R2	0.19	0.47
AIC	230.2	249.6
Controls	Yes	Yes
Time FEs	No	Yes
N	637	637

Ordered probit. Robust standard errors. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Though these estimates might appear small, they are economically significant, considering the likelihood of having a dissenting vote. To put this into perspective, in our sample, only 6% of cases members were dissents, with 4% of hawkish and 2% dovish dissents. One standard deviation increase in *Hawk* doubles the likelihood of hawkish dissent. Thus, the estimated effects of the *Hawk* measure are substantial, demonstrating the measure’s relevance for explaining dissenting votes. These results are qualitatively similar to the findings of Istrefi (2019), though obtained with completely different data and framework.

4.4 Strategic forecasting in FOMC

If members were engaged in strategic forecasting, the change in voting status would change the incentives to predict according to consensus. For example, a non-voting member might be willing to steer the consensus by submitting abnormally low or high forecasts. When a member gains voting status, such incentives should become weaker, as voting members can influence the policy outcome directly through their vote. This is a potential reason why the forecasted federal funds rates would not imply the preferred rates, which could result in failing to capture the true preferences of FOMC members.

To test the extent of strategic forecasting in the sample, I look at how voting status changes affect the forecasts, relative to consensus. I estimate the following regression

on the sample of regional Fed presidents:

$$|\mathbf{E}_t^i X_{t+h} - \mathbf{E}_t^{mean} X_{t+h}| = \alpha_i + \alpha_t + \beta_1 NonV_t^i + \epsilon_t,$$

where $\mathbf{E}_t^i X_{t+h}$ denotes the forecast of one of the three: *{inflation, unemployment rate, federal funds rate}*, $\mathbf{E}_t^{mean} X_{t+h}$ is the cross-sectional mean of forecasts across all voting members except i , and $NonV_t^i = 1$ if member i is a non-voter in period t , and zero otherwise. I restrict the sample to members who change their voting status over the sample period. By examining the coefficient β_1 , we can assess whether members alter their forecasting behavior when their voting status changes.²⁵

Tables 3 and 4 report the estimated coefficients for inflation and unemployment, respectively. In all cases, the Non-Vote dummy variable coefficients are relatively close to zero. The effects are not statistically significant at the 5% level for all except two cases. Thus, non-voting status does not appear to result in economically meaningful deviations from the consensus forecast inputs. Importantly, Table 5 similarly shows that non-voting members do not systematically deviate from the consensus federal funds rate forecast relative to voting members. Therefore, voting status should not substantially affect the estimates of average marginal effects. Thus, I do not observe systematic strategic forecasting in the sample.²⁶

Table 3: Effect of Non-Voting Status on **Inflation Rate** Forecasts

	$h = 1$	$h = 2$	$h = 3$	$h = LR$
β_1	0.004 (0.013)	0.001 (0.014)	0.015 (0.015)	-0.001 (0.005)
<i>Adj.R</i> ²	0.62	0.70	0.66	0.84
N	552	552	552	447
Robust standard errors in parentheses				
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$				

²⁵Note that the use of absolute deviations focuses on the magnitude of strategic forecasts rather than their direction.

²⁶Alternative approaches exist to test for strategic forecasting behavior (e.g., using narrative data or survey metadata), but these are beyond the scope of this paper.

Table 4: Effect of Non-Voting Status on **Unemployment Rate** Forecasts

	$h = 1$	$h = 2$	$h = 3$	$h = LR$
β_1	0.011 (0.008)	0.033** (0.013)	0.028* (0.017)	0.020 (0.013)
$Adj.R^2$	0.62	0.66	0.65	0.80
N	552	552	552	443

Robust standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Effect of Non-Voting Status on **Federal funds Rate** Forecasts

	$h = 1$	$h = 2$	$h = 3$	$h = LR$
β_1	0.017 (0.013)	0.039 (0.027)	0.061 (0.045)	-0.030 (0.018)
$Adj.R^2$	0.66	0.81	0.82	0.73
N	348	348	348	267

Robust standard errors in parentheses
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 Conclusion

In this paper, I use Summary of Economic Projections (SEP) data to estimate FOMC member-level policy reaction functions for 2012–2019. I merge individual SEP forecasts with participant keys (released with a five-year lag) to construct a panel of policymakers’ forecasts, and I estimate a Tobit model for expectations-based reaction functions of 23 members.

I document substantial heterogeneity in estimated reaction functions across FOMC members. For example, a 1-percentage-point increase in the forecasted unemployment gap is associated with a change in the forecasted federal funds rate ranging from -1.9 to 0 percentage points, while a comparable increase in the forecasted inflation gap is associated with a change ranging from 0 to 3.4 percentage points. I document that this heterogeneity in reaction functions implies tangible variation in forecasted policy rates. I also show that this high level of heterogeneity in inflation and unemployment gap coefficients can be attributed to the position of the member, educational background,

and work and research experience. PhD holders in economics and related fields put relatively less weight on inflation compared to holders of other degrees (MBA, BA, etc.). Board members tend to put less weight on inflation, while members with publications in top-5 economic journals tend to put more weight on it. Older members and those with longer Federal Reserve experience place more weight on economic activity, as measured by the unemployment gap.

To summarize heterogeneity in reaction functions, I construct a parsimonious measure, *Hawk*, for the analyzed sample of FOMC members, based on the inflation and unemployment gap average marginal effects. This measure represents how the inflation-unemployment tradeoff is perceived across FOMC members. To further demonstrate the importance of reaction function heterogeneity, I show that dissent voting is associated with the estimated measure of the inflation-unemployment tradeoff: it is much more likely to vote for tighter (looser) policy when a member is estimated to be more hawkish (dovish).

A key limitation of my approach hinges on the structural features of the SEP panel. The average FOMC member serves about a five-year consecutive term on the committee, while participants' IDs are subject to considerable publication lags, which prevent tracking FOMC members' responsiveness to inflation and economic activity in real time. However, the availability and transparency of SEP data continue to improve, and it will be possible to obtain more precise estimates of individual policy reaction functions and extend the sample of members further in the future. Given the growing calls for transparency, it would not be surprising if the Federal Reserve eventually reduces or eliminates the publication lag entirely.

Taken together, our findings offer empirical evidence of heterogeneity within FOMC reaction functions. The analysis suggests that this heterogeneity might stem from different preferences or different beliefs of FOMC members about data-generating processes in the economy. There are, however, other potential sources of variation in reaction functions. Clear identification of these sources appears to be an important avenue for future research.

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